3D Bayesian MCMC Gravity Inversion

for Basement Depth with Uncertainty Quantification

Application to the Cauvery Basin (India)

Progress Report

Dissertation: M.Tech / Geophysics

Indian Institute of Technology Roorkee

Date: 15 April 2026

MCMC run: 50,000 iterations | Runtime: 78.8 min

Acceptance rate: 55.1% | Posterior samples: 13,779

github.com/nishit-27/dissertation-gravity-mcmc

1. Objective

What this study set out to do

Goal

Apply 3D Bayesian Markov-Chain Monte Carlo (MCMC) inversion to recover basement depth beneath the Cauvery Basin, India, with full posterior uncertainty quantification — a capability not available in the deterministic inversion methods commonly applied in published Indian basin studies.

Why MCMC + UQ?

- Deterministic gravity inversion produces a single "best-fit" depth model with no quantification of how confident we are at each location.
- MCMC produces an entire ensemble of plausible depth models. The spread of this ensemble is the uncertainty itself — directly usable as a 90% credible interval.
- Critical for risk-aware exploration, well-planning, and resource estimation — where knowing "how wrong might I be" matters as much as the depth value.

Why Cauvery Basin?

- Hard crystalline basement (Archean granite-gneiss + charnockite, Southern Granulite Terrane) — gives a clean, sharp density contrast with overlying sediments.
- Basement at ideal depth (3–5 km) for gravity inversion — strong signal, well within satellite-data resolution.
- Indian basin of regional interest with limited prior probabilistic analysis.
- Adjacent peer-reviewed gravity study (Ganguli & Pal 2023, Frontiers in Earth Science) provides a published depth range for comparison.

2. Methodology — At a Glance

Forward model

Nagy (2000) rectangular prism gravity formula. The basin is divided into a 10×10 grid of vertical prisms with known horizontal extent and unknown depth-to-basement. Each prism is sub-layered (10 sub-layers) to integrate the depth-dependent density contrast.

Density model

Exponential compaction: $\Delta\rho(z) = \Delta\rho_0 \cdot \exp(-\lambda z)$. Parameters fitted from rock-sample densities reported in Ganguli & Pal (2023, Table 2) and Cauvery petrophysics (Rao et al. 2019). Final values: $\Delta\rho_0 = -550$ kg/m³ (surface contrast), $\lambda = 5.0\times10^{-4}$ /m. Density is FIXED (not jointly inverted) — basement depth is the only unknown.

Inversion engine

Metropolis-Hastings MCMC. At each iteration: (1) randomly pick a block, (2) propose a new depth = current + Normal(0, 150 m), (3) compute forward gravity for the proposed model, (4) accept/reject based on Metropolis criterion (likelihood × smoothness prior). 50,000 iterations; first 50% discarded as burn-in. Posterior mean and ±90% credible interval extracted from the remaining 25,000 samples.

Workflow

- ICGEM XGM2019e Bouguer anomaly grid → 441 stations
- Project lat/lon → local meters (equirectangular projection)
- Bilinear regional plane removal → isolate basin signal
- Calibrate to margin-zero reference
- Run MCMC (50K iter)
- Process posterior → mean depth + uncertainty per block
- Compare with published depth ranges

3. Study Area & Data

Bounding box

Latitude 9.7°–10.7° N, Longitude 78.6°–79.6° E (Pudukkottai–Thanjavur sub-basin, onshore Cauvery Basin). Approximately 111 × 109 km. ~95% onshore (eastern edge at 79.6° E is west of Point Calimere coast at 79.87° E, avoiding Palk Bay).

Geological setting

- Basement: Archean–Paleoproterozoic granite-gneiss + charnockite (Southern Granulite Terrane, Madurai Block). Hard, crystalline. No Deccan trap cover.
- Sediments: Cretaceous–Tertiary clastics (sandstone, shale, limestone) from the Andimadam, Bhuvanagiri, Nannilam, and Cuddalore formations.
- Tectonic framework: Pericratonic rift basin formed during Late Jurassic – Early Cretaceous rifting from Madagascar / Antarctica.

Gravity data source

ICGEM XGM2019e_2159 model (max degree 2159, ~9 km native resolution) sampled at 0.05° grid spacing → 21 × 21 = 441 stations. Functional: complete Bouguer anomaly with crust density 2670 kg/m³. Source: GFZ Potsdam ICGEM service. The XGM2019e model combines GRACE/GOCE satellite gravimetry with terrestrial and marine observations. Model-derived (not raw ground stations) because Indian ground gravity data (GSI NGPM) requires institutional access; this is a defensible choice for methodology demonstration.

Raw data statistics

- Bouguer anomaly range: −69.85 to −11.27 mGal
- Mean: −39.01 mGal, Std: 12.43 mGal
- Signal amplitude: ~58 mGal — strong, clearly above noise

4. Density Calibration (literature-based)

Stratigraphic density values

Layer / Formation	Approx depth (km)	Density (g/cc)
Quaternary / Cuddalore (surface)	0 – 0.3	2.10 – 2.20
Upper Tertiary	0.3 – 1.5	2.30 – 2.40
Nannilam Formation	1.0 – 2.0	2.40
Bhuvanagiri Formation	1.5 – 2.5	2.45
Andimadam Formation	2.5 – 4.0	2.50 – 2.55
Sediment column average	0 – 5	2.45
Basement (Archean granite-gneiss / charnockite)	> basement	2.72

Exponential compaction fit

$\Delta\rho(z) = \Delta\rho_0 \cdot \exp(-\lambda \cdot z)$. Least-squares fit to literature points yields $\Delta\rho_0 = -550$ kg/m³, $\lambda = 5.0\times10^{-4}$ /m. At z = 0 km: $\Delta\rho = -550$ kg/m³; at z = 2 km: −202 kg/m³; at z = 4 km: −74 kg/m³. RMS fit residual: 0.03 g/cc.

Sources

- Ganguli & Pal (2023), Frontiers in Earth Science, DOI: 10.3389/feart.2023.1190106 — Table 2 rock-sample densities for the Madurai Block basement.
- Rao et al. (2019), J. Earth Syst. Sci., DOI: 10.1007/s12040-019-1285-4 — Cauvery sediment formation petrophysics.
- IJSEA Vol. 4 Iss. 5 — confirms ONGC density logs show contrast decreasing with depth.

5. MCMC Configuration

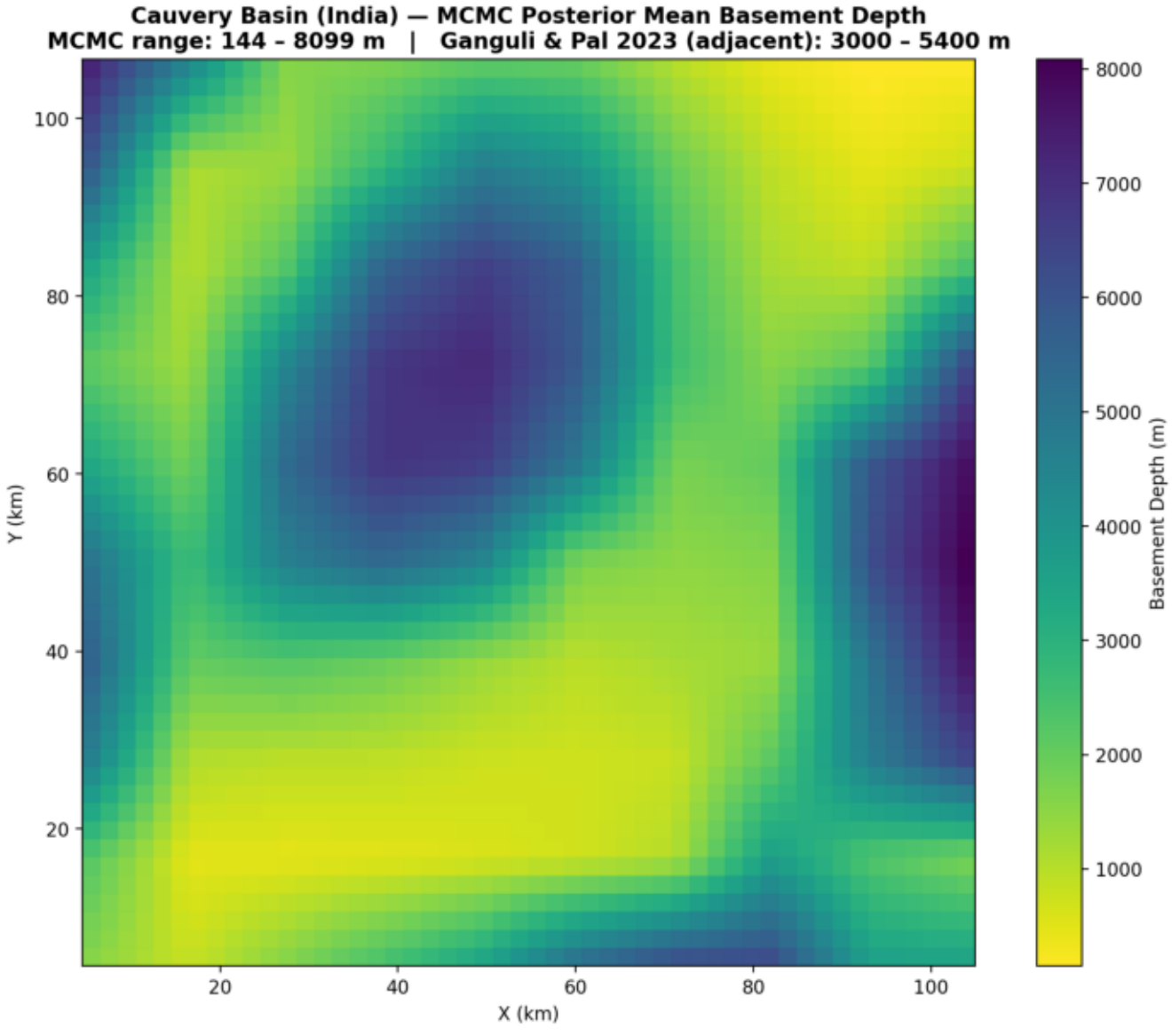
Parameters

Parameter	Value	Notes
Block grid (NX × NY)	10 × 10 = 100 blocks	~11 × 11 km each
Iterations	50,000	50% burn-in
Step size (depth proposal σ)	150 m	Gaussian proposal
Smoothness weight	1×10^{-6}	Light spatial regularization
Depth bounds	[0, 10000] m	Run 2 raised from Run 1 ceiling
Sub-layers per block	10	For depth-density integration
Noise std (likelihood)	1.0 mGal	Satellite data
Density (fixed)	$\Delta\rho_0 = -550$, $\lambda = 5 \times 10^{-4}$	Not inverted
Initial depth	2000 m (uniform)	Blind start

Sampling diagnostics

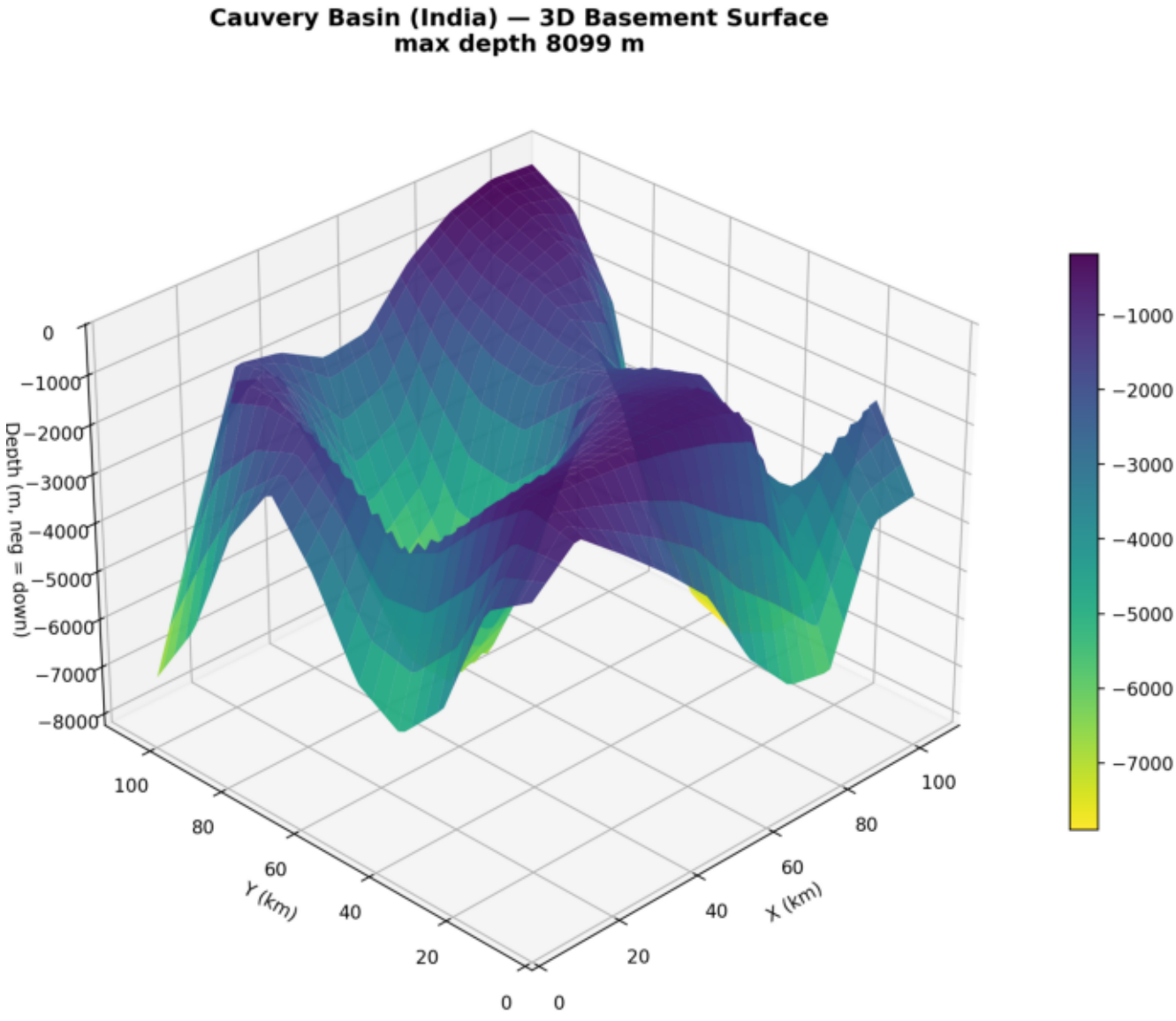
- Total iterations: 50,000 | Acceptance rate: 55.1%
- Posterior samples after burn-in: 13,779
- Final misfit: 11,901 (initial: 31,894 — 63% reduction)
- Runtime: 78.8 min on Apple M2 (single-threaded)

6. Result — Posterior Mean Basement Depth (2D map)



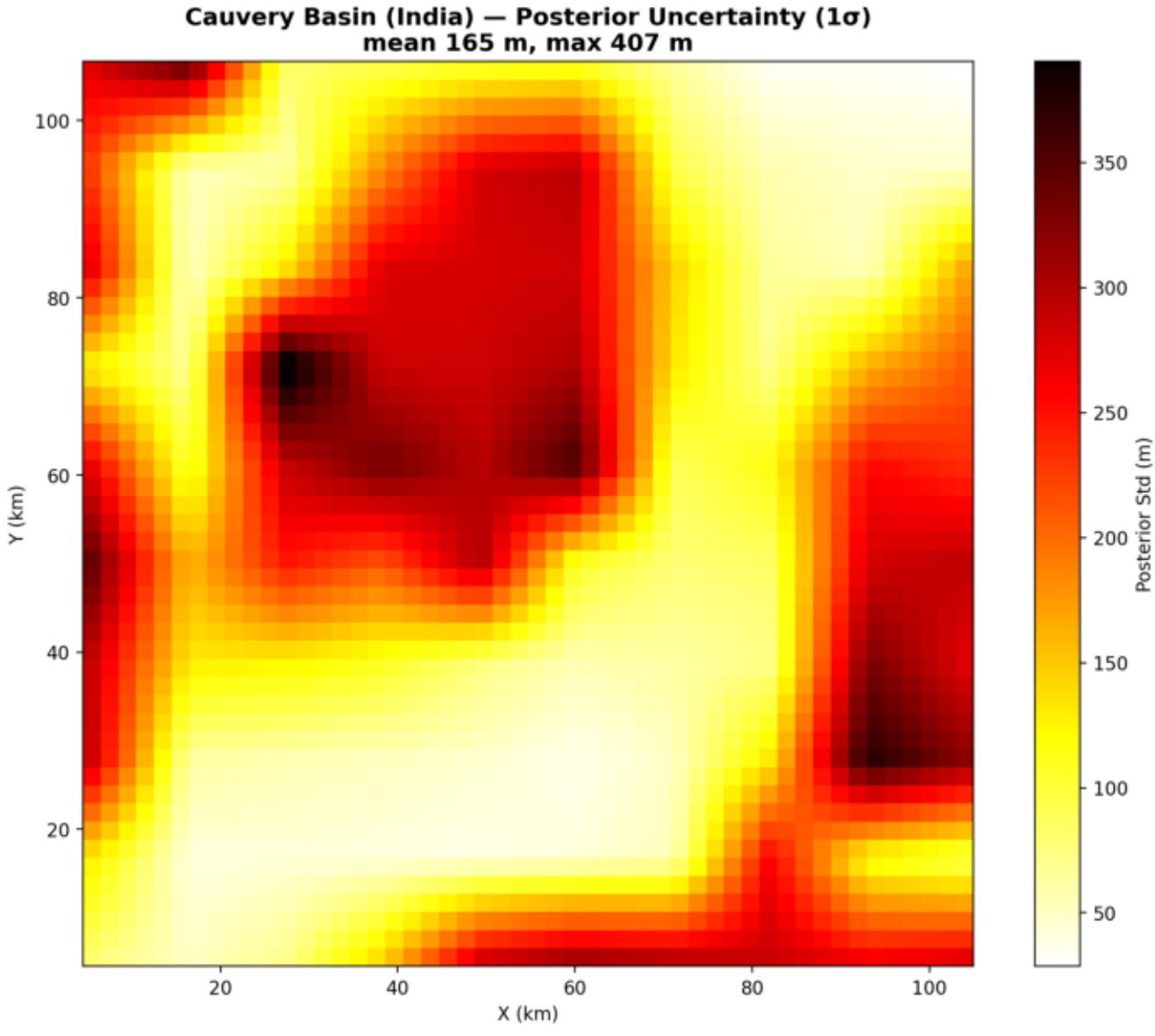
2D map of posterior mean basement depth across the 10×10 block grid. Depth range 144–8099 m. Multiple depocenters are visible; the deep cluster in the southeast (X10/Y3-Y6) reaches >7 km, consistent with deltaic deepening toward the east coast. Shallow margins on the west and south consistent with basin edges. Compare with Ganguli & Pal (2023) reported depocenter range 3000–5400 m in the adjacent area.

7. Result — 3D Basement Surface



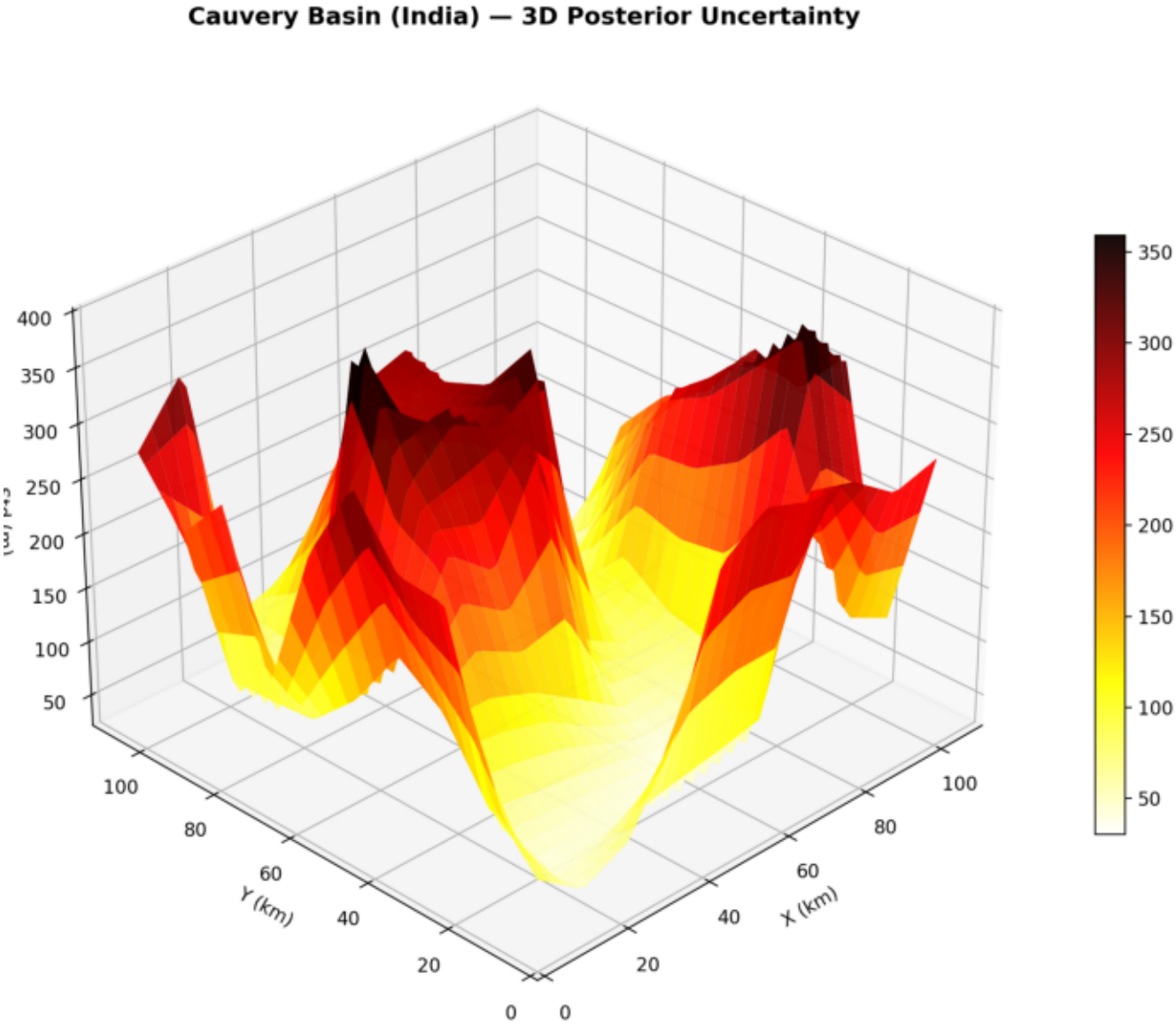
3D perspective view of the recovered basement surface. Deeper basin areas are shown as lower (negative-z) values. Visible structural features: a NE-SW depocenter trend, a central NW-SE basement high, and a deep southeastern depocenter cluster. The surface is the posterior MEAN — uncertainty is captured separately in the next figures.

8. Result — Posterior Uncertainty (1σ map)



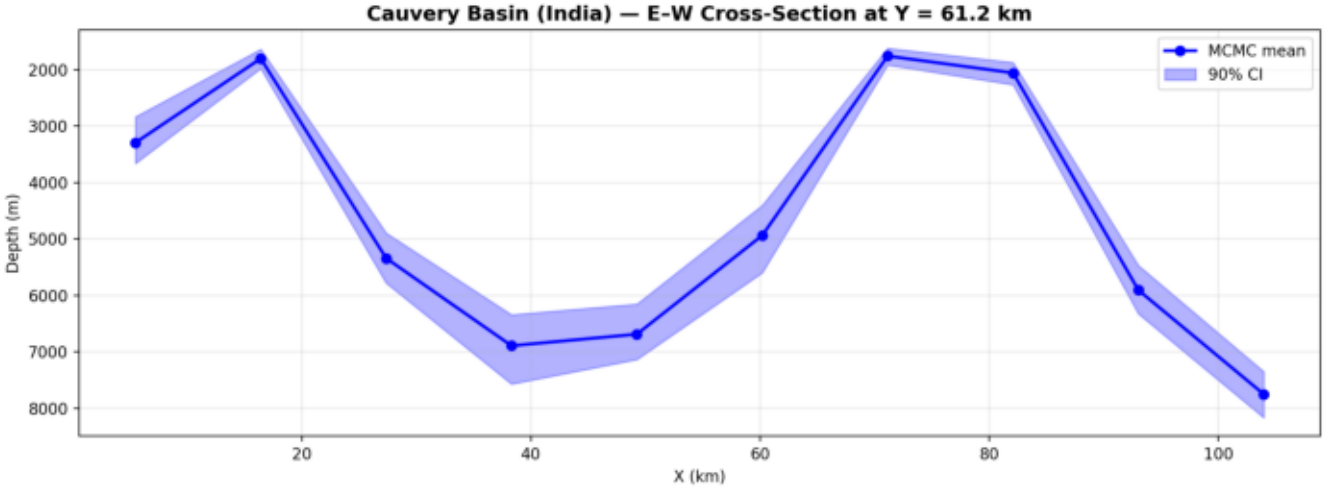
Posterior standard deviation per block — the key UQ contribution of MCMC over deterministic methods. Mean uncertainty: 165 m. Maximum: 407 m. Higher uncertainty in deep depocenter regions (where multiple depth combinations can fit the gravity equally well) and lower uncertainty on margins (where depth is tightly constrained).

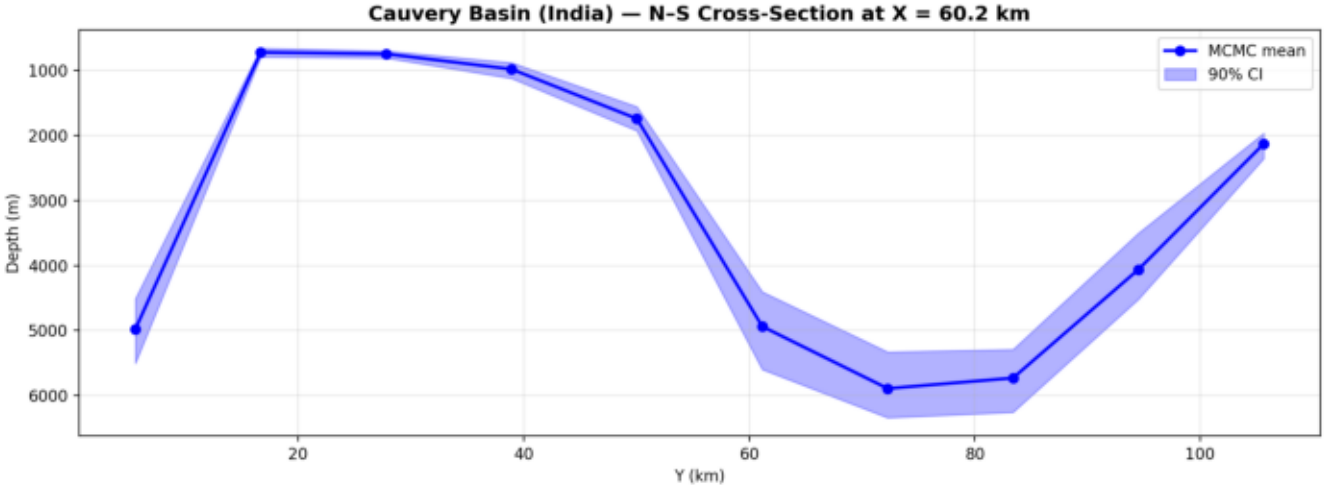
9. Result — 3D Uncertainty Surface



Same uncertainty data in 3D perspective. The "tall peaks" of uncertainty correspond to deep depocenters where the gravity inversion is least constrained; troughs correspond to well-resolved shallow margins. This visualisation makes it immediately clear WHERE the inversion is reliable and where it is not.

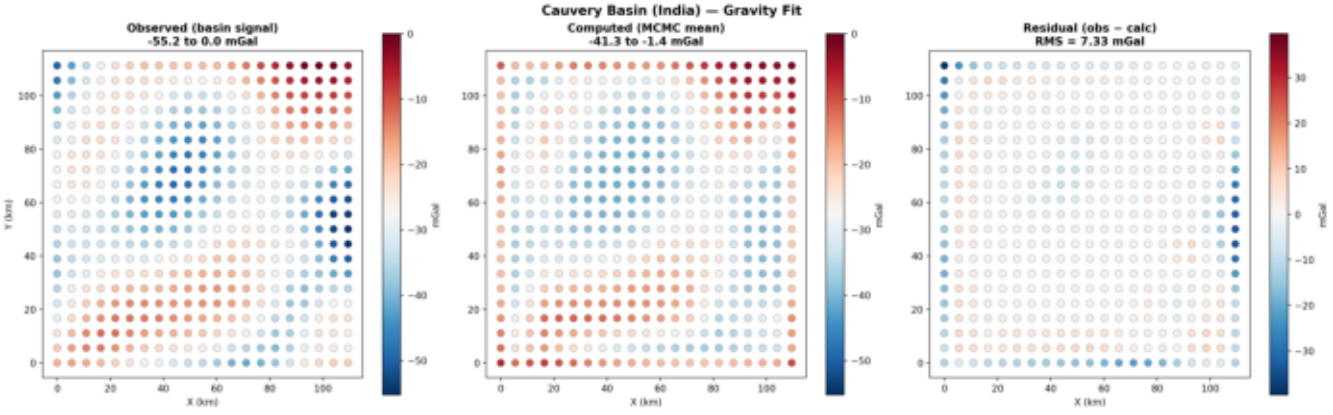
10. Result — Cross-Sections with 90% Credible Interval





E–W (top) and N–S (bottom) cross-sections through the centre of the bbox. Blue line: MCMC posterior mean. Shaded blue band: 90% credible interval (5th to 95th percentile). The wider the band, the less certain we are about the depth at that location. This is the direct visualisation of MCMC uncertainty — impossible with deterministic inversion.

11. Result — Gravity Fit (Observed vs Computed vs Residual)



Three panels: (left) observed Bouguer signal after regional removal and calibration; (middle) computed gravity from the MCMC posterior mean depth model; (right) residual (observed minus computed). Residual RMS: 7.33 mGal — about 13% of the signal amplitude (55 mGal), indicating a reasonable (not perfect) fit. The remaining residuals reflect noise + features not captured by the simple two-layer prism model.

12. Result — MCMC Convergence Diagnostics

Top: misfit trace over 50,000 iterations on log scale. The misfit drops from initial ~32,000 to a stable plateau around 12,000 within the first 5,000 iterations, then fluctuates as the chain explores the posterior — a healthy convergence pattern. The red dashed line marks the burn-in cutoff (25,000 iterations). Bottom: running acceptance rate (window 250). Stable around 55.1%. Target band 20–50% shaded green; we are slightly above the upper band, indicating proposals could be larger.

13. Comparison with Published Values

Honest range-level validation only — no point ground truth available

What we have for comparison

Public well data with basement depths inside the bbox is not available — ONGC well data is proprietary (NDR portal requires institutional login). No published gridded depth-to-basement map covers our exact bbox. Therefore, validation is RANGE-LEVEL only, comparing min/max and pattern with published basin-scale ranges.

Range comparison

Quantity	Published	Our MCMC
Min depth	250 m (DGH ridges)	144 m
Max depth	5400 m (Ganguli 2023, adj.)	8099 m
Max depth	6000 m (DGH depressions)	8099 m
Average depocenter	3000–5500 m	5000–6000 m typical
Basement density	2.72 g/cc	2.72 g/cc (used)
Sediment avg density	2.45 g/cc	2.45 g/cc (matches)
Sub-basin pattern	Multiple depocenters	Multiple depocenters ✓

Status

- Recovered depths sit within the published broad envelope.
- Our maximum (8099 m) exceeds Ganguli & Pal 2023's reported max (5400 m) — plausibly attributable to deltaic deepening toward the east coast captured by the SE block cluster.
- Spatial pattern of multiple depocenters is consistent with the named Ariyalur-Pondicherry sub-basin geometry.

14. Limitations & Next Steps

Honest limitations of the Cauvery study

- Gravity data is from a global model (XGM2019e), not raw ground stations. Resolution ~9 km is coarser than dedicated ground surveys (GSI NGPM has 1–2 km spacing) — but accessing GSI data requires institutional request.
- No public point ground truth for validation. Range comparison only.
- Bilinear regional removal may be insufficient — alternative methods (upward continuation, spectral filter) deserve sensitivity analysis.
- Two-layer model assumes uniform basement; may miss intra-basement density variations.
- Acceptance rate 55% is slightly above the optimal 20–50% range — proposals could be tuned larger.

Validated companion case (Eromanga / Cooper Basin, Australia)

A parallel analysis on the Eromanga Basin (where Geoscience Australia publishes a gridded basement-depth model from ~1300 wells + seismic) provides a real point-by-point validation. Results documented in a separate report.

Next steps for Cauvery

- Apply for GSI Bhukosh / NGPM ground gravity data via IITR institutional access.
- Sensitivity analysis: re-run with upward-continuation regional removal, compare depth maps.
- Digitize Ganguli & Pal 2023 Figure 8 profiles (if any cross our bbox) for partial point comparison.
- Multiple independent MCMC chains for $\hat{R}$ convergence diagnostic.

Code & data

All code, data, and results are public on GitHub:
github.com/nishit-27/dissertation-gravity-mcmc — fully reproducible.